

Software Requirements Specification Document

BLOOD BANK

Group - 7

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1. Project Summary

1.1. Project Overview

The Blood Bank Management System is a web-based application designed to streamline and manage blood donation, request, and inventory processes. The system provides a platform for donors, patients, hospitals, and administrators to interact in a coordinated and efficient manner. Developed using HTML, Tailwind CSS, and JavaScript, the website features a visually appealing, responsive layout, and intuitive user experience. The application enables users to register as donors or request blood, while hospitals can register to coordinate blood-related logistics. The system also provides a login interface and a live blood inventory viewer.

1.2. Project Scope

The Blood Bank Management System includes essential features for various stakeholders:

- **Donor Features:** Donors can register their details including name, blood group, contact, donation date, and quantity.
- **Patient Features:** Patients can request blood by entering details like blood group, date of requirement, and contact info.
- **Hospital Features:** Hospitals can register with contact details to be a part of the donation and request process.
- **Blood Inventory View:** A real-time inventory display showing available blood groups and their quantities.
- **Admin/Login Interface:** A login interface (placeholder) that can be extended to support authentication and admin dashboards.

- **UI/UX:** Visually enriched, GIF-integrated pages with user-friendly forms, mobile responsiveness, and themed consistency using Tailwind CSS.
- **Constraints:** There are no current limitations on data storage or user authentication, but future scalability for advanced analytics and machine learning integration is planned.

1. General Description

1.1 Product Functions Overview

The Blood Bank Management System offers the following functional modules:

- **Home Page:** Features a full-screen GIF banner with the slogan “Save Lives, Donate Blood” and quick navigation.
- **Donor Registration Page:** Input form for donor ID, age, blood group, donation date, quantity, and contact.
- **Patient Request Page:** Allows patients to request blood by entering ID, name, blood group, date, and location.
- **Hospital Registration Page:** Form to register hospitals with name, ID, city, state, and contact number.
- **Blood Inventory Page:** Tabular display of available blood groups and corresponding quantities.
- **Login Page:** A simple login form for future admin or hospital user integration.
- **Navigation:** Present on all pages with links to Home, Donor, Patient, Inventory, Hospital, and Login.
- **Footer:** Present on most pages with consistent branding and contact information.

2.2 Types of Users

1. Donor:

Registers their details and makes blood donations available for inventory.

2. Patient:

Submits blood requests specifying the group and date of need.

3. Hospital Representative:

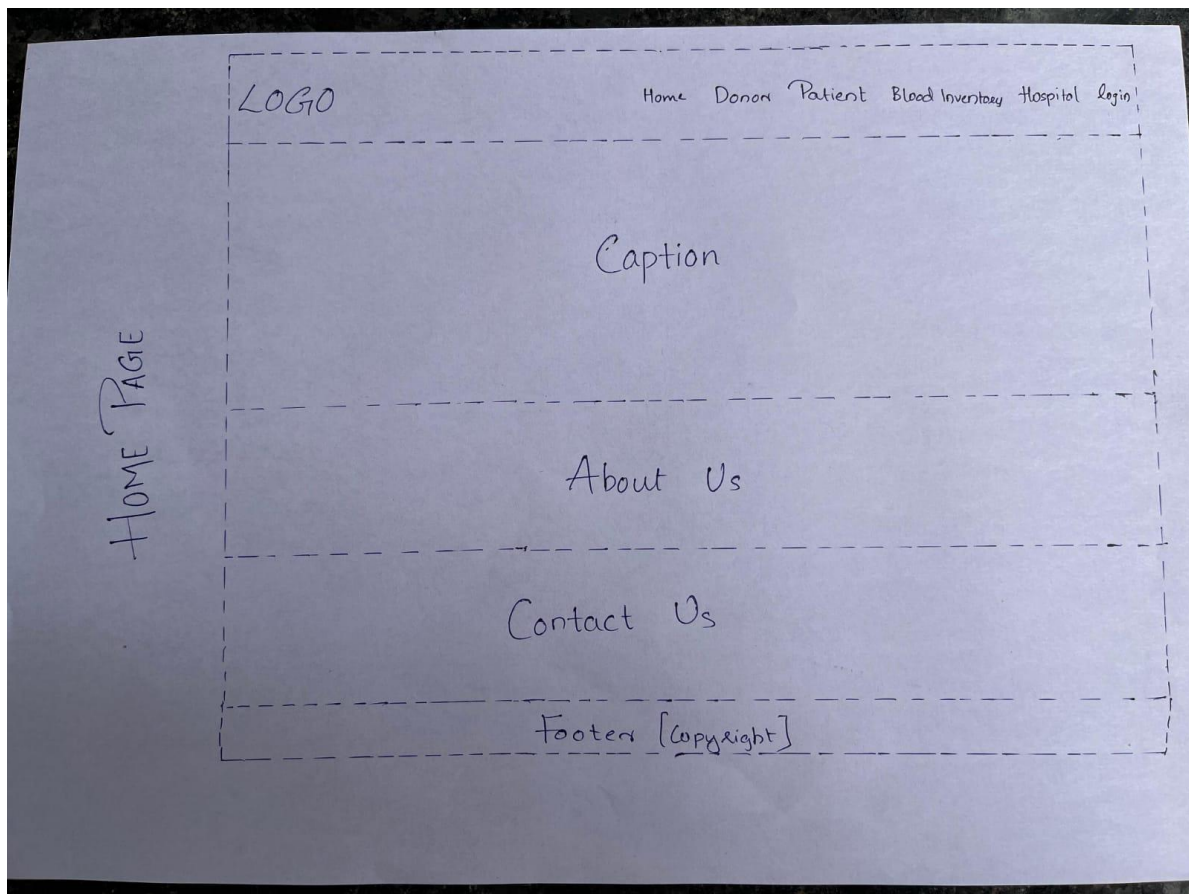
Registers hospital details for system access and coordination.

3. Technical Requirements

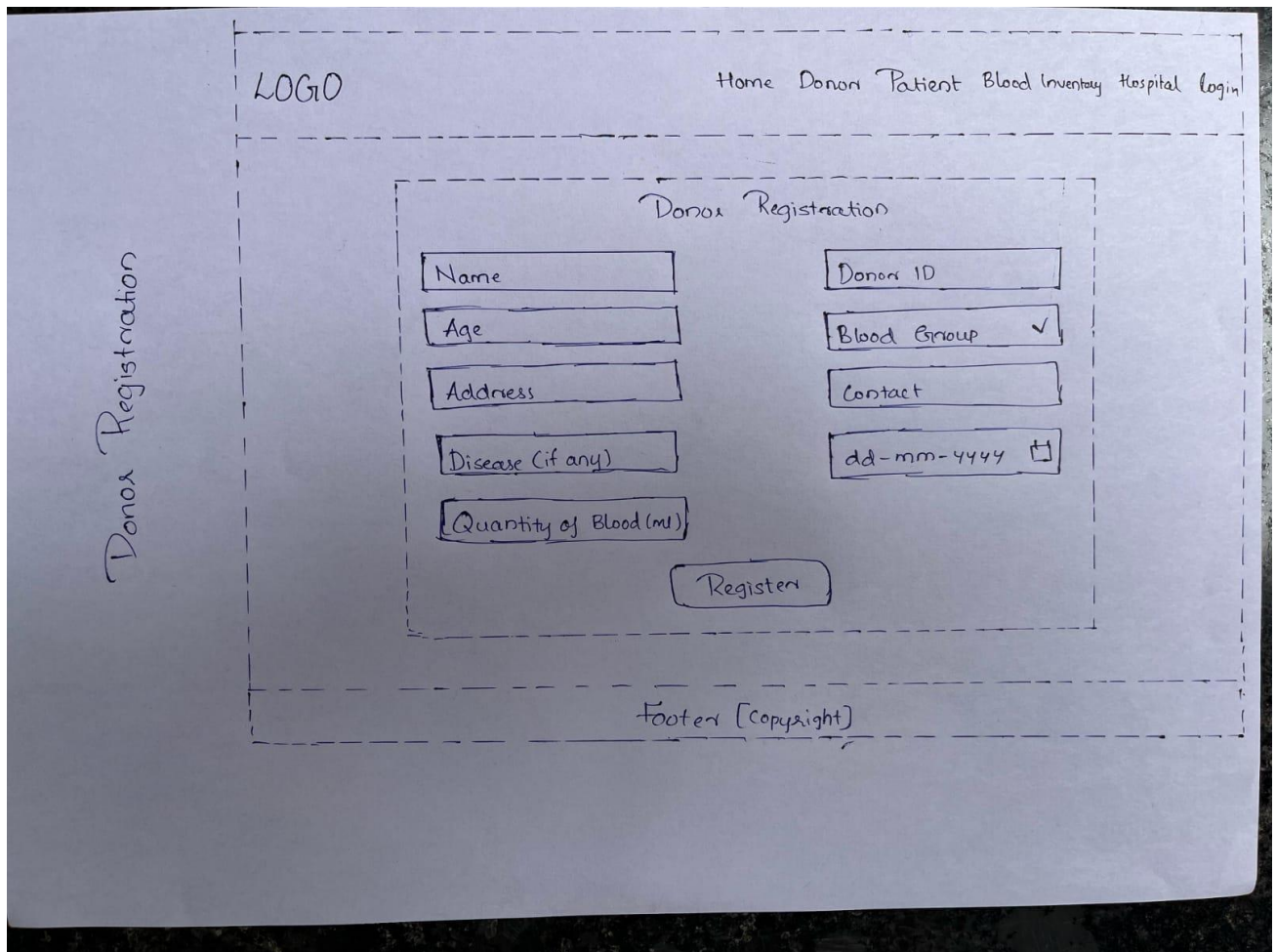
- **Frontend:** React Js, Tailwind CSS, Node Js
- **Backend :** PostgreSQL + Node.js
- **Architecture:** The system follows a **full-stack architecture**, integrating a dynamic HTML frontend styled with Tailwind CSS and a backend that handles authentication, form submission and database interactions using PostgreSQL

- **Design:** Mobile-first responsive UI with GIF backgrounds, theme colors (red, white, soft grey)
- **Animations:** CSS-based animations like slide-up effect; visual appeal through background GIFs
- **Data:** A relational database.
- **Browser Compatibility:** Chrome, Firefox, Safari, Edge (latest versions).

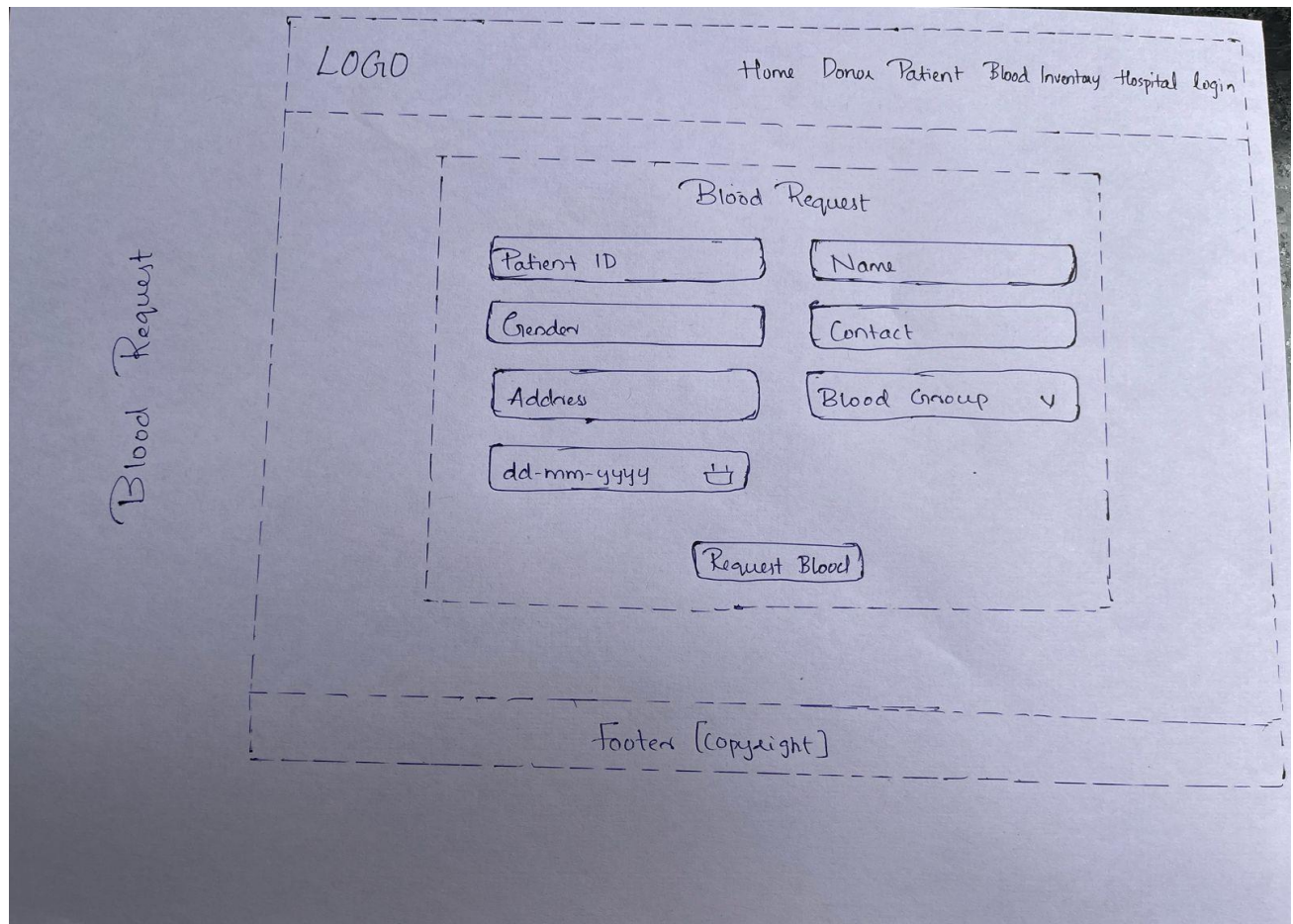
4. User Interface (visual layout of the elements)



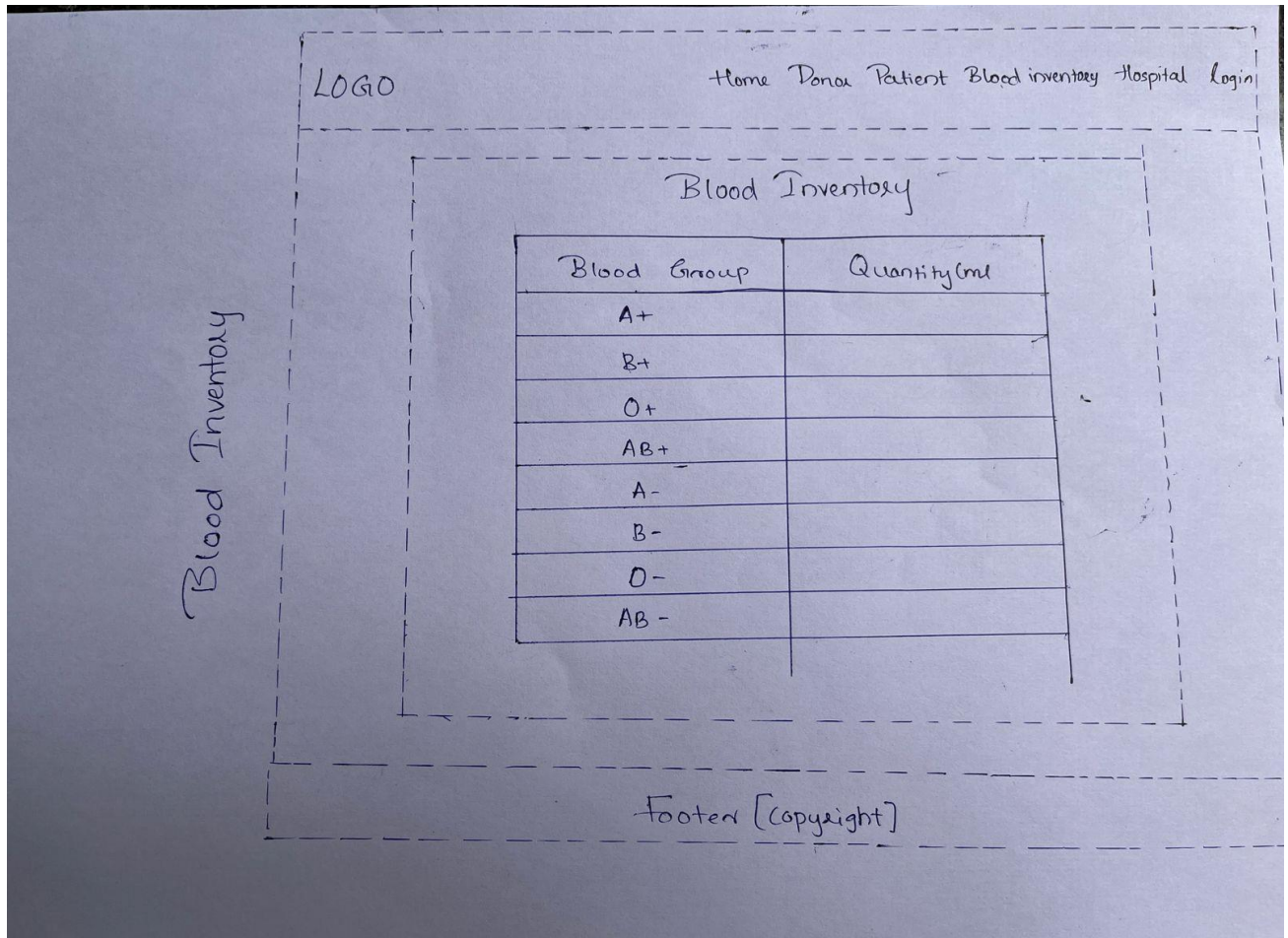
Home Page UI



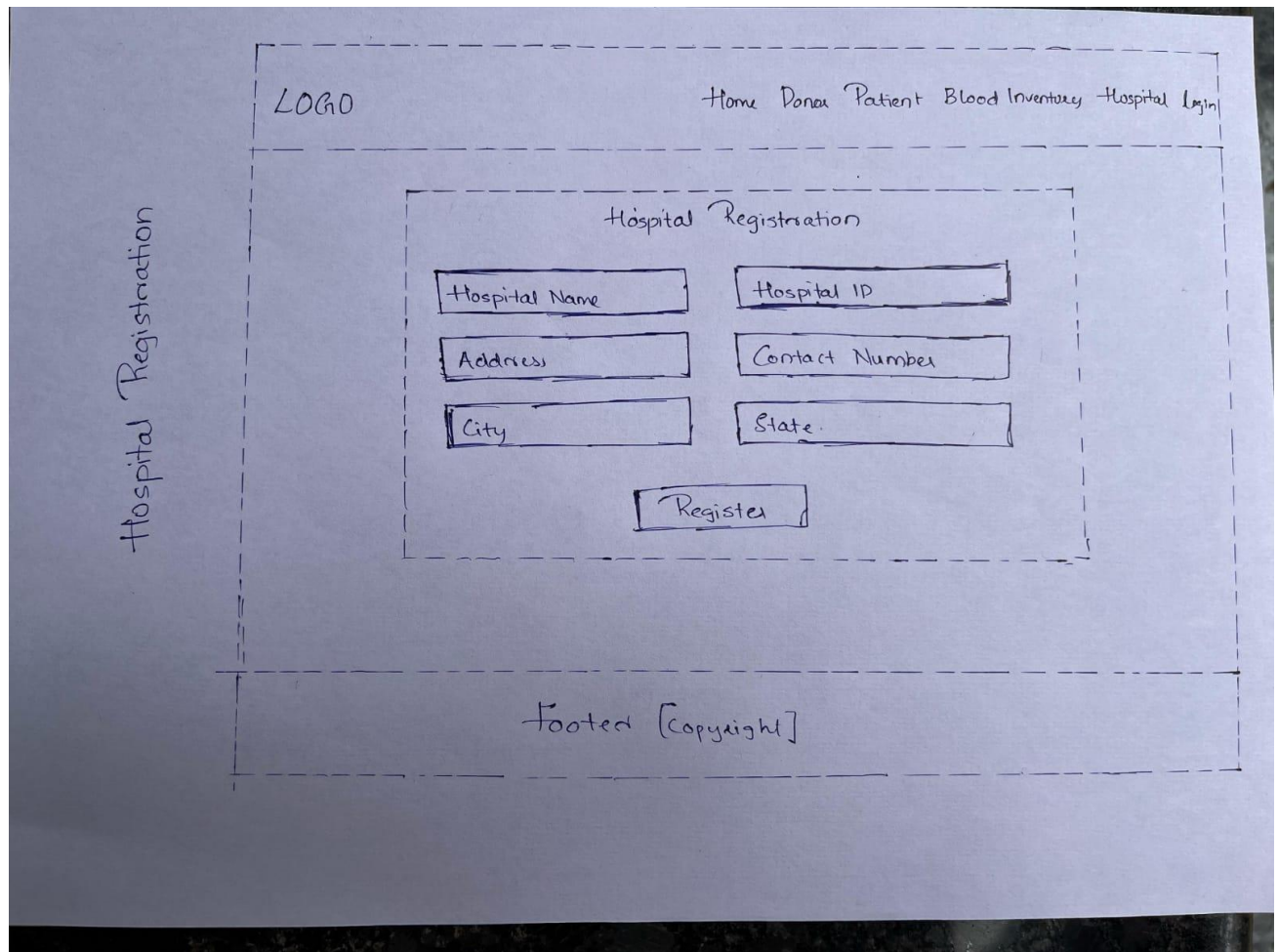
Donor Registration UI



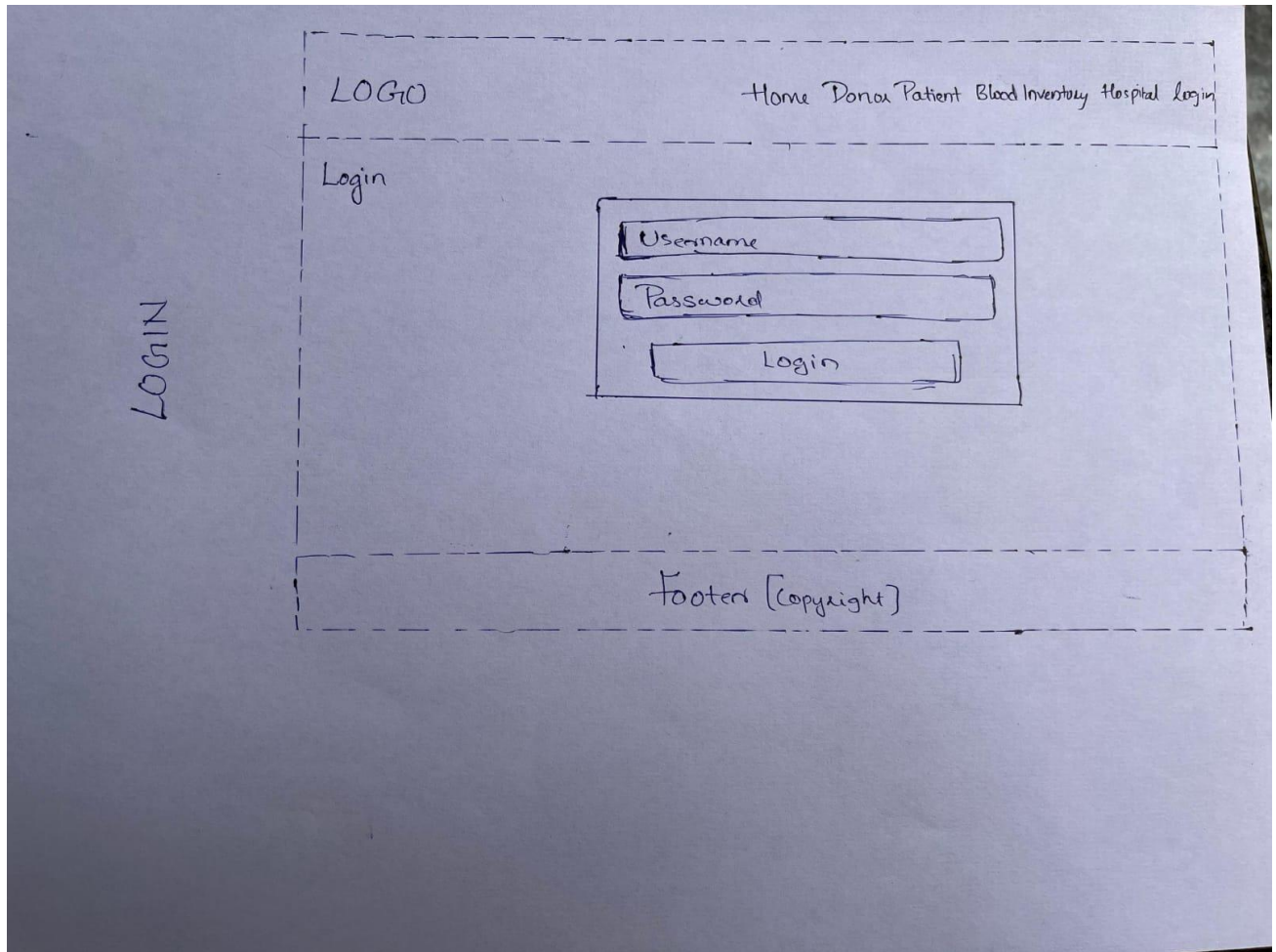
Blood Request UI



Blood Inventory UI



Hospital Registration UI



Login Page UI

Key Screens:

- **Home Page:**
 - Fullscreen hero GIF with headline and slogan
 - “About Us” and “Contact Us” sections with contact details
- **Donor Registration Page:**
 - Multi-field form with inputs for ID, name, age, blood group, quantity, date, and contact
- **Patient Request Page:**
 - Form with dropdown for blood group, contact information, and required date
- **Hospital Registration Page:**
 - Clean UI form with fields for hospital name, ID, city, and contact
- **Blood Inventory Page:**
 - Table layout showing current stock of blood by group and quantity
- **Login Page:**
 - Form with username and password fields
 - Decorative GIF for visual engagement
- **Navigation Bar:**
 - Appears on all pages with links to every section for easy access
- **Footer:**
 - Present on the homepage and registration pages with copyright

Team members and their Contribution:

Team Member	Feature Ownership
Shishira Prakash	Blood Inventory Module, Database Schema Design (PostgreSQL)
Amitha K Anilkumar	UI Design and Animations
Gopika V	Homepage, Navigation Bar, Footer, Hospital Registration
Nithya Anvitha	Donor and Patient Registration Forms, Data Validation, Backend Integration